

PART A — QUESTIONS

AF7-001

Question:

Solve: $9x + 4 = 58$

- A) 5
- B) 6
- C) 7
- D) 8

AF7-002

Question:

If $7n - 15 = 34$, what is n ?

- A) 5
- B) 6
- C) 7
- D) 8

AF7-003

Question:

Simplify: $2(6x + 5) - 3x$

- A) $9x + 10$
- B) $12x + 10$
- C) $9x + 5$
- D) $15x + 10$

AF7-004

Question:

A rental company charges \$18 per hour plus a \$12 flat fee. Which expression represents the total cost for h hours?

- A) $18h + 12$
- B) $12h + 18$
- C) $30h$
- D) $18 - 12h$

AF7-005

Question:

If $f(x) = 7x - 9$, what is $f(5)$?

- A) 26
- B) 28
- C) 35
- D) 44

AF7-006

Question:

Which ordered pair satisfies the equation $y = 2x - 5$?

- A) (3, 2)
- B) (4, 3)
- C) (5, 6)
- D) (6, 9)

AF7-007

Question:

Solve: $8(x - 4) = 40$

- A) 7
- B) 8
- C) 9
- D) 10

AF7-008

Question:

A pattern begins: 9, 15, 21, 27, ... What is the next number?

- A) 31
- B) 32
- C) 33
- D) 34

AF7-009

Question:

Solve: $x/12 = 4$

- A) 16
- B) 24
- C) 36
- D) 48

AF7-010

Question:

A line has a slope of 5 and crosses the y-axis at -2. Which equation represents the line?

- A) $y = -2x + 5$
- B) $y = 5x - 2$
- C) $y = 2x - 5$
- D) $y = -5x + 2$

AF7-011

Question:

If 4 crates contain 68 parts, how many parts are in 7 crates at the same rate?

- A) 105
- B) 112
- C) 119
- D) 126

AF7-012

Question:

Solve: $36 - 6x = 12$

- A) 3
- B) 4
- C) 5
- D) 6

AF7-013

Question:

Which expression is equivalent to $21c - 8c + 6$?

- A) $13c + 6$
- B) $29c + 6$
- C) $13c - 6$
- D) $21c - 2$

AF7-014

Question:

A storage room starts with 75 boxes. Workers add 10 boxes each hour. Which equation gives the total number of boxes B after h hours?

- A) $B = 75h + 10$
- B) $B = 10h + 75$
- C) $B = 85h$
- D) $B = 75 - 10h$

AF7-015

Question:

If $y = -6x + 20$, what is y when $x = 2$?

- A) 8
- B) 12
- C) 22
- D) 32

AF7-016

Question:

Solve: $12x - 8 = 5x + 34$

- A) 5
- B) 6
- C) 7
- D) 8

AF7-017

Question:

Which value of x makes $4x + 7$ less than 31?

- A) 5
- B) 6
- C) 7
- D) 8

AF7-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : -5, 2, 9, 16

What is the rule?

- A) $y = 7x - 5$
- B) $y = 5x - 7$
- C) $y = -5x + 7$
- D) $y = 7x + 5$

AF7-019

Question:

Solve: $3(x + 8) = x + 38$

- A) 5
- B) 6
- C) 7
- D) 8

AF7-020

Question:

If $a = -7$ and $b = 5$, what is $2a + 3b$?

- A) -1
- B) 1
- C) 15
- D) 29

AF7-021

Question:

Which point has a negative x-coordinate and a positive y-coordinate?

- A) (4, -6)
- B) (-4, 6)
- C) (-4, -6)
- D) (4, 6)

AF7-022

Question:

A worker checks 56 items in 7 minutes. At the same rate, how many items can the worker check in 11 minutes?

- A) 77
- B) 84
- C) 88
- D) 96

AF7-023

Question:

Solve for x: $P = 9x - 18$

- A) $x = P + 18$
- B) $x = (P + 18)/9$
- C) $x = 9(P + 18)$
- D) $x = P/9 - 18$

AF7-024

Question:

What is the slope of a line that goes through (2, 1) and (8, 19)?

- A) 2
- B) 3
- C) 4
- D) 6

AF7-025

Question:

Simplify: $15x - 11 - 7x + 4$

- A) $8x - 7$
- B) $22x - 7$
- C) $8x - 15$
- D) $22x + 15$

AF7-026

Question:

If $g(x) = x^2 - 4x$, what is $g(8)$?

- A) 24
- B) 32
- C) 48
- D) 64

AF7-027

Question:

A sequence follows the rule “subtract 5, then multiply by 2.” If the first term is 14, what is the second term?

- A) 9
- B) 18
- C) 23
- D) 28

AF7-028

Question:

Solve: $7(x + 2) - 5 = 44$

- A) 5
- B) 6
- C) 7
- D) 8

AF7-029

Question:

A line crosses the y-axis at 3 and falls 4 units for every 1 unit to the right. Which equation represents the line?

- A) $y = 4x + 3$
- B) $y = -4x + 3$
- C) $y = 3x - 4$
- D) $y = -3x + 4$

AF7-030

Question:

If $9/13 = x/39$, what is x ?

- A) 18
- B) 24
- C) 27
- D) 30

AF7-031

Question:

Solve: $-9x + 20 = -43$

- A) -7
- B) -5
- C) 5
- D) 7

AF7-032

Question:

A container starts with 450 nails. Each project uses 30 nails. Which expression gives the number of nails left after p projects?

- A) $450 + 30p$
- B) $450 - 30p$
- C) $30p - 450$
- D) $450 \div 30p$

AF7-033

Question:

Which equation has $x = 11$ as its solution?

- A) $2x + 7 = 31$
- B) $4x - 9 = 35$
- C) $x/11 + 6 = 8$
- D) $5x - 3 = 50$

PART B — ANSWER KEY + EXPLANATIONS

AF7-001 — Correct: B — Explanation:

Solve $9x + 4 = 58$. Subtract 4 from both sides: $9x = 54$. Divide by 9: $x = 6$. The most common mistake is subtracting 4 correctly but forgetting to divide by 9.

AF7-002 — Correct: C — Explanation:

Start with $7n - 15 = 34$. Add 15 to both sides: $7n = 49$. Divide by 7: $n = 7$. A common mistake is subtracting 15 instead of adding it.

AF7-003 — Correct: A — Explanation:

Distribute first: $2(6x + 5) = 12x + 10$. Then subtract $3x$: $12x - 3x + 10 = 9x + 10$. A common mistake is forgetting to multiply 5 by 2.

AF7-004 — Correct: A — Explanation:

The company charges \$18 for each hour, so that part is $18h$. The \$12 flat fee is added one time. The total cost is $18h + 12$. A common mistake is treating the flat fee as the hourly rate.

AF7-005 — Correct: A — Explanation:

Substitute 5 for x : $f(5) = 7(5) - 9 = 35 - 9 = 26$. A common mistake is adding 9 instead of subtracting it.

AF7-006 — Correct: B — Explanation:

Test the choices in $y = 2x - 5$. For $(4, 3)$, $2(4) - 5 = 8 - 5 = 3$, which matches the y -value. A common mistake is choosing a point that looks close but does not make the equation true.

AF7-007 — Correct: C — Explanation:

Solve $8(x - 4) = 40$. Divide both sides by 8: $x - 4 = 5$. Add 4: $x = 9$. A common mistake is adding 4 before undoing the multiplication by 8.

AF7-008 — Correct: C — Explanation:

The pattern increases by 6 each time: 9, 15, 21, 27. Add 6 to 27: 33. A common mistake is assuming the pattern increases by 5 because the numbers are close together.

AF7-009 — Correct: D — Explanation:

$x/12 = 4$ means x divided by 12 equals 4. Multiply both sides by 12: $x = 48$. A common mistake is adding 12 and 4 instead of multiplying.

AF7-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is 5 and the y-intercept is -2, so the equation is $y = 5x - 2$. A common mistake is switching the slope and y-intercept.

AF7-011 — Correct: C — Explanation:

Find the unit rate: $68 \text{ parts} \div 4 \text{ crates} = 17 \text{ parts per crate}$. For 7 crates: $17 \times 7 = 119$. A common mistake is adding 3 crates and 3 parts instead of using the rate.

AF7-012 — Correct: B — Explanation:

Solve $36 - 6x = 12$. Subtract 36 from both sides: $-6x = -24$. Divide by -6: $x = 4$. A common mistake is losing the negative sign after subtracting 36.

AF7-013 — Correct: A — Explanation:

Combine like terms: $21c - 8c = 13c$. The constant 6 stays the same. The expression becomes $13c + 6$. A common mistake is combining the constant with the variable terms.

AF7-014 — Correct: B — Explanation:

The storage room starts with 75 boxes. Workers add 10 boxes each hour. After h hours, the total is $B = 10h + 75$. A common mistake is making 75 the hourly rate.

AF7-015 — Correct: A — Explanation:

Substitute $x = 2$ into $y = -6x + 20$: $y = -6(2) + 20 = -12 + 20 = 8$. A common mistake is treating $-6(2)$ as positive 12.

AF7-016 — Correct: B — Explanation:

Solve $12x - 8 = 5x + 34$. Subtract $5x$: $7x - 8 = 34$. Add 8: $7x = 42$. Divide by 7: $x = 6$. A common mistake is adding 8 to only one side.

AF7-017 — Correct: A — Explanation:

Solve $4x + 7 < 31$. Subtract 7: $4x < 24$. Divide by 4: $x < 6$. The only listed value less than 6 is 5. A common mistake is choosing 6, but $4(6) + 7 = 31$, not less than 31.

AF7-018 — Correct: A — Explanation:

The y-values increase by 7 each time x increases by 1, so the slope is 7. When $x = 0$, $y = -5$, so the y-intercept is -5. The rule is $y = 7x - 5$. A common mistake is using +5 instead of -5.

AF7-019 — Correct: C — Explanation:

Distribute: $3x + 24 = x + 38$. Subtract x : $2x + 24 = 38$. Subtract 24: $2x = 14$. Divide by 2: $x = 7$. A common mistake is forgetting to multiply 8 by 3.

AF7-020 — Correct: B — Explanation:

Substitute $a = -7$ and $b = 5$: $2a + 3b = 2(-7) + 3(5) = -14 + 15 = 1$. A common mistake is treating $2(-7)$ as positive 14.

AF7-021 — Correct: B — Explanation:

A negative x-coordinate and a positive y-coordinate means the point is in Quadrant II. The point $(-4, 6)$ fits this description. A common mistake is choosing $(4, -6)$, which has the signs reversed.

AF7-022 — Correct: C — Explanation:

56 items in 7 minutes means $56 \div 7 = 8$ items per minute. In 11 minutes: $8 \times 11 = 88$. A common mistake is multiplying 56 by 11 without finding the rate first.

AF7-023 — Correct: B — Explanation:

Start with $P = 9x - 18$. Add 18 to both sides: $P + 18 = 9x$. Divide by 9: $x = (P + 18)/9$. A common mistake is dividing P by 9 first and then adding 18.

AF7-024 — Correct: B — Explanation:

Slope equals change in y divided by change in x. From $(2, 1)$ to $(8, 19)$, y increases by 18 and x increases by 6. The slope is $18 \div 6 = 3$. A common mistake is using 18 as the slope without dividing by the change in x.

AF7-025 — Correct: A — Explanation:

Combine like terms: $15x - 7x = 8x$, and $-11 + 4 = -7$. The simplified expression is $8x - 7$. A common mistake is writing $8x + 7$ because the negative sign before 11 is missed.

AF7-026 — Correct: B — Explanation:

Substitute 8 for x: $g(8) = 8^2 - 4(8) = 64 - 32 = 32$. A common mistake is calculating 8^2 as 16 instead of 64.

AF7-027 — Correct: B — Explanation:

Start with 14. Subtract 5: 9. Then multiply by 2: 18. A common mistake is multiplying first and then subtracting 5.

AF7-028 — Correct: A — Explanation:

Solve $7(x + 2) - 5 = 44$. Distribute: $7x + 14 - 5 = 44$. Combine: $7x + 9 = 44$. Subtract 9: $7x = 35$. Divide by 7: $x = 5$. A common mistake is forgetting to combine $14 - 5$.

AF7-029 — Correct: B — Explanation:

“Falls 4 units for every 1 unit to the right” means the slope is -4 . The line crosses the y-axis at 3, so the equation is $y = -4x + 3$. A common mistake is using a positive slope because the number 4 is given.

AF7-030 — Correct: C — Explanation:

Use the proportion $9/13 = x/39$. Since $13 \times 3 = 39$, multiply 9 by 3: $x = 27$. A common mistake is adding 26 to the numerator because 13 becomes 39.

AF7-031 — Correct: D — Explanation:

Solve $-9x + 20 = -43$. Subtract 20 from both sides: $-9x = -63$. Divide by -9 : $x = 7$. A common mistake is giving -7 because both sides contain negative numbers.

AF7-032 — Correct: B — Explanation:

The container starts with 450 nails. Each project uses 30 nails, so p projects use $30p$ nails. The number left is $450 - 30p$. A common mistake is adding $30p$ even though nails are being used.

AF7-033 — Correct: B — Explanation:

Test $x = 11$. In choice B, $4(11) - 9 = 44 - 9 = 35$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.